

we tend to think that what happens every decade in fact only happens once every century, and, furthermore, that we know what's going on.

This miscalculation problem is a little more subtle. In truth, outliers are not as sensitive to underestimation since they are fragile to estimation errors, which can go in both directions. As we saw in [Chapter 6](#), there are conditions under which people overestimate the unusual or some specific unusual event (say when sensational images come to their minds)—which, we have seen, is how insurance companies thrive. So my general point is that these events are very fragile to *miscalculation*, with a general severe underestimation mixed with an occasional severe overestimation.

The errors get worse with the degree of remoteness to the event. So far, we have only considered a 2 percent error rate in the game we saw earlier, but if you look at, say, situations where the odds are one in a hundred, one in a thousand, or one in a million, then the errors become monstrous. The longer the odds, the larger the epistemic arrogance.

Note here one particularity of our intuitive judgment: even if we lived in Mediocristan, in which large events are rare (and, mostly, inconsequential), we would still underestimate extremes—we would think that they are even rarer. We underestimate our error rate even with Gaussian variables. Our intuitions are sub-Mediocristani. But we do not live in Mediocristan. The numbers we are likely to estimate on a daily basis belong largely in Extremistan, i.e., they are run by concentration and subjected to Black Swans.

Guessing and Predicting

There is no effective difference between my guessing a variable that is not random, but for which my information is partial or deficient, such as the number of lovers who transited through the bed of Catherine II of Russia, and predicting a random one, like tomorrow's unemployment rate or next year's stock market. In this sense, guessing (what I don't know, but what someone else may know) and predicting (what has not taken place yet) are the same thing.

To further appreciate the connection between guessing and predicting, assume that instead of trying to gauge the number of lovers of Catherine of Russia, you are estimating the less interesting but, for some, more important question of the population growth for the next century, the stockmarket returns, the social-security deficit, the price of oil, the results of your great-uncle's estate sale, or the environmental conditions of Brazil two decades from now. Or, if you are the publisher of Yevgenia Krasnova's book, you may need to produce an estimate of the possible future sales. We are now getting